Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The value of $\log ab - \log |b| =$

(A) $\log a$

(B) $\log |a|$

(C) $-\log a$

(D) none of these

Q.2

If $a, b, c \in R^+$, then $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$ is always

(A) ≥ 12

(B) ≥ 9

(C) ≤ 12

(D) none of these

Q.3

If $f(x) = \log\left(\frac{1+x}{1-x}\right)$, then

(A) $f(x_1) \cdot f(x_2) = f(x_1 + x_2)$

(B) $f(x+2) - 2f(x+1) + f(x) = 0$

(C) $f(x) + f(x+1) = f\left(x^2 + x\right)$

(D) $f(x_1) + f(x_2) = f\left(\frac{x_1+x_2}{1+x_1x_2}\right)$

Q.4

The solution set of $\log_2 |4 - 5x| > 2$ is

(A) $\left(\frac{8}{5}, \infty\right)$

(B) $\left(\frac{4}{5}, \frac{8}{5}\right)$

(C) $(-\infty, 0) \cup \left(\frac{8}{5}, \infty\right)$

(D) none of these

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Q.5

If $\log_y x + \log_x y = 2, x^2 + y = 12$, then the value of xy is

(A) 9

(B) 12

(C) 15

(D) 21

Q.6

The solution of the inequality $\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$ is

(A) $[1, 3] \cup (5, \infty)$

(B) $(1, 3) \cup (5, \infty)$

(C) $(-\infty, 1) \cup (5, \infty)$

(D) none of these

Q.7

The value of $\frac{1+2\log_3 2}{(1+\log_3 2)^2} + (\log_6 2)^2$ is

(A) 2

(B) 3

(C) 4

(D) 1

Q.8

If x satisfies $|x-1| + |x-2| + |x-3| \geq 6$, then

(A) $0 \leq x \leq 4$

(B) $x \leq -2$ or $x \geq 4$

(C) $x \leq 0$ or $x \geq 4$

(D) none of these

Q.9

The number of roots of the equation $\log_{3\sqrt{x}} x + \log_{3x} \sqrt{x} = 0$ is

(A) 1

(B) 2

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(C) 3

(D) 0

Q.10

The solution set of $\frac{x^2-3x+4}{x+1} > 1, x \in R$

(A) $(3, \infty)$

(B) $(-1, 1) \cup (3, \infty)$

(C) $[-1, 1] \cup [3, \infty)$

(D) none of these

Q.11

The values of x, which satisfy both the inequations $x^2 - 1 \leq 0$ and $x^2 - x - 2 \geq 0$ lie in

(A) $(-1, 2)$

(B) $\{-1, 1\}$

(C) $\{1, 2\}$

(D) $\{-1\}$

Q.12

If $\log_4 5 = a$ and $\log_5 6 = b$, then $\log_3 2$ is equal to

(A) $\frac{1}{2a+1}$

(B) $\frac{1}{2b+1}$

(C) $2ab + 1$

(D) $\frac{1}{2ab-1}$

Q.13

The value of $3^{\log_4 5} - 5^{\log_4 3}$ is

(A) 0

(B) 1

(C) 2

(D) none of these

Q.14

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The solution set of the inequality $\log_{0.8} \log_6 \left(\frac{x^2+x}{x+4}\right) < 0$ is

(A) $[-4,-2] \cup (8,\infty)$

(B) $(-4,-3) \cup [8,\infty)$

(C) $(-4,-3) \cup (8,\infty)$

(D) none of these

Q.15

If $y = 3^{x-1} + 3^{-x-1}$, then the least value of y is

(A) 2

(B) 6

(C) $2/3$

(D) $3/2$

Q.16

The set of all possible real values of a such that the inequality $(x-(a-1))\left(x-\left(a^2+2\right)\right) < 0$ holds for all $x \in (-1,3)$ is

(A) $(0,1)$

(B) $(\infty,-1]$

(C) $(-\infty,-1]$

(D) $(1,\infty)$

Q.17

If $xy^2 = 4$ and $\log_3 (\log_2 x) + \log_{1/3}\left(\log_{1/2} y\right) = 1$, then x equals

(A) 4

(B) 8

(C) 16

(D) 64

Q.18

If $(21.4)^a = (0.00214)^b = 100$, then the value of $\frac{1}{a} - \frac{1}{b}$ is

(A) 0

(B) 1

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(C) 2

(D) 4

Q.19

If $\log_2 x + \log_2 y \geq 6$, then the least value of $x + y$ is

(A) 4

(B) 8

(C) 16

(D) 32

Q.20

Complete set of values of x satisfying inequality $||x-1|-5| < 2x-5$ is

(A) $(5/2, \infty)$

(B) $(11/3, \infty)$

(C) $(-1, \infty)$

(D) $(-\infty 1/3)$

Q.21

The value of $\frac{\log_2 24}{\log_{96} 2} - \frac{\log_2 192}{\log_{12} 2}$ is

Q.22

If $\log_3 \{5 + 4\log_3(x-1)\} = 2$, then x is equal to

Q.23

The value of x satisfying $\sqrt{3}^{-4+2\log_{\sqrt{5}} x} = 1/9$ is

Q.24

If $\log_2 x + \log_x 2 = \frac{10}{3} = \log_2 y + \log_y 2$, then $x + y =$

Q.25

The number $N = 6\log_{10} 2 + \log_{10} 31$ lies between two successive integers whose sum is equal to

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Q.1 Answer:

$\log |a|$

Solution:

$= \log ab - \log |b|$
$= \log |a| + \log |b| - \log |b|$
$= \log |a|.$

Q.2 Answer:

≥ 9

Solution:

Using A.M and H.M inequality we get
$\frac{(a+b+c)}{3} \geq \frac{3}{\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)} (\because \text{ A. } M \geq H.M)$
$\Rightarrow (a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$

Q.3 Answer:

$f(x_1) + f(x_2) = f\left(\frac{x_1+x_2}{1+x_1x_2}\right)$

Solution:

$$f(x_1) + f(x_2) = \log\left(\frac{1+x_1}{1-x_1} \cdot \frac{1+x_2}{1-x_2}\right)$$
$$= \log\left(\frac{1+x_1x_2+x_1+x_2}{1+x_1x_2-x_1-x_2}\right)$$
$$= \log\left(\frac{1+\frac{x_1+x_2}{1+x_1x_2}}{1-\frac{x_1+x_2}{1+x_1x_2}}\right) = f\left(\frac{x_1+x_2}{1+x_1x_2}\right)$$

Q.4 Answer:

$(-\infty, 0) \cup \left(\frac{8}{5}, \infty\right)$

Solution:

$\log_2 |4-5x| > 2 \Rightarrow |4-5x| > 2^2 = 4 \Rightarrow \left|\frac{5x}{4} - 1\right| > 1 \Rightarrow \frac{5x}{4} - 1 > 1$
or $\frac{5x}{4} - 1 < -1 \Rightarrow x > \frac{8}{5}$ or $x < 0 \Rightarrow x \in (-\infty, 0) \cup \left(\frac{8}{5}, \infty\right)$

Q.5 Answer:

9

Solution:

We have $\log_y x + \log_x y = 2, x^2 + y = 12$
If $\log_y x = a$**, then** $\log_x y = \frac{1}{a}, \left[\because \log_a b = \frac{1}{\log_b a}\right]$
$\Rightarrow \mathrm{a} + \frac{1}{\mathrm{a}} = 2$

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$\Rightarrow a^2 + 1 = 2a$

$\Rightarrow (a-1)^2 = 0$

$\Rightarrow a = 1$

Hence, $y = x$

Substituting in the equation, we get

$x^2 + x - 12 = 0$

Hence, $x = 3$ or $x = -4$

Since x is a base in the logarithmic function, x has to be positive. Hence, $x = 3$ and $y = 3$

Therefore, $xy = 9$

Q.6 Answer:

$[1, 3] \cup (5, \infty)$

Solution:

$\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$

$\frac{2x+14+(x-5)(3x+1)}{2(x-5)} \geq 0$

$\frac{2x+14+3x^2+x-15x-5}{(x-5)} \geq 0$

$\frac{3x^2-12x+9}{(x-5)} \geq 0$

$\frac{x^2-4x+3}{(x-5)} \geq 0$

$\frac{x^2-3x-x+3}{(x-5)} \geq 0$

$\frac{(x-3)(x-1)}{(x-5)} \geq 0 \quad 0 \quad \left(x = 4, \quad \frac{1\times 3}{-1} < 0\right)$

$x \varepsilon [1, 3] \cup [5, \infty)$

Q.7 Answer:

1

Solution:

$N = \frac{1+2\log_3 2}{(1+\log_3 2)^2} + \log_6^2 2$

$= \frac{1+2\log_3 2}{(1+\log_3 2)^2} + \frac{1}{(\log_2 6)^2} \quad \left[\because \log_a b = \frac{1}{\log_b a}\right]$

$= \frac{1+2\log_3 2}{(1+\log_3 2)^2} + \frac{1}{\{\log_2(2\times 3)\}^2}$

$= \frac{1+2\log_3 2}{(1+\log_3 2)^2} + \frac{1}{\left(\log_2^2 + \log_2 3\right)^2}$

$= \frac{1+2\log_3 2}{(1+\log_3 2)^2} + \frac{1}{(1+\log_2 3)^2} \left[\because \log_a a = 1\right]$

$= \frac{1+2\log_3 2}{\left(1+\log_3^2\right)^2} + \frac{1}{(1+\log_2 3)^2} \times \frac{(\log_3 2)^2}{(\log_3 2)^2}$

$= \frac{1+2\log_3 2}{(1+\log_3 2)^2} + \frac{(\log_3 2)^2}{\left(\log_3 2 + \log_2{}^3 \log_3^2\right)^2}$

$= \frac{1+2\log_3 2}{(1+\log_3 2)^2} + \frac{(\log_3 2)^2}{\log_3 2 + 1} \quad \left[\because \log_a b = \frac{1}{\log_b a}\right]$

$= \frac{1+(\log_3 2)^2+2(1)(\log_3 2)}{(1+\log_3 2)^2}$

$= \frac{(1+\log_3 2)^2}{(1+\log_3 2)^2} = 1$ which is a pime number

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Q.8 Answer:

$x \leq 0$ or $x \geq 4$

Solution:

For $x \leq 1$, the given inequation becomes

$1 - x + 2 - x + 3 - x \geq 6 \Rightarrow -3x \geq 0 \quad \Rightarrow x \leq 0$

and for $x \geq 3$, the given equation becomes

$x - 1 + x - 2 + x - 3 \geq 6 \Rightarrow 3x \geq 12 \Rightarrow x \geq 4$

For $1 < x \leq 2$ we get $x - 1 + 2 - x + 3 - x \geq 6 \Rightarrow -x + 4 \geq 6$ i.e., $-x \geq 2 \Rightarrow x \leq -2$ not possible.

For $2 < x < 3$, we get $x - 1 + x - 2 + 3 - x \geq 6 \Rightarrow x \geq 6$, not possible. Hence solution set is $(-\infty, 0] \cup [4, \infty)$

i.e. $x \leq 0$ or $x \geq 4$

Q.9 Answer:

2

Solution:

$\frac{\log x}{\log 3 + \frac{1}{2}\log x} + \frac{\frac{1}{2}\log x}{\log 3 + \log x}$

$\Rightarrow \frac{\log_3 x}{1 + \frac{1}{2}\log_3 x} + \frac{1}{2}\frac{\log_3 x}{(1 + \log_3 x)} = 0$

Let $\log_3 x = y$, we get $\frac{y}{1 + \frac{y}{2}} + \frac{y}{2(1+y)} = 0.$

$\Rightarrow y[4 + 4y + 2 + y] = 0$

$\Rightarrow y = 0$ **or** $y = -\frac{6}{5}$

$\Rightarrow \log_3 x = 0$ **or** $\log_3 x = -\frac{6}{5}$

$\Rightarrow x = 1$ **or** $x = 3^{-\frac{6}{5}}$

Q.10 Answer:

$(-1, 1) \cup (3, \infty)$

Solution:

Given,

$\frac{x^2 - 3x + 4}{x + 1} > 1$

$\Rightarrow \frac{x^2 - 3x + 4}{x + 1} - 1 > 0$

$\Rightarrow \frac{x^2 - 3x + 4 - x - 1}{x + 1} > 0 \Rightarrow \frac{x^2 - 4x + 3}{x + 1} > 0$

$\Rightarrow \frac{x^2 - 3x - x + 3}{x + 1} > 0 \Rightarrow \frac{x(x-3) - 1(x-3)}{x + 1} > 0$

$\Rightarrow \frac{(x-3)(x-1)}{x+1} > 0 \Rightarrow (x - 3)(x - 1)(x + 1) > 0$

$\Rightarrow x \in (-1, 1) \cup (3, \infty)$

Q.11 Answer:

$\{-1\}$

Solution:

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$x^2 - 1 \leq 0 \Rightarrow \quad (x+1)(x-1) \leq 0 \quad \Rightarrow -1 \leq x \leq 1$

$x^2 - x - 2 \geq 0 \quad \Rightarrow (x+1)(x-2) \geq 0 \Rightarrow x \leq -1 \text{ or } x \geq 2$

$\therefore$ **Both in equations hold only when** $x = -1$

Q.12 Answer:

$\frac{1}{2ab-1}$

Solution:

We have,

$\log_4 5 = \text{a and } \log_5 6 = \text{b}$

Now using, $\log_y x = \frac{\log x}{\log y}$**, we get**

$a = \frac{\log 5}{\log 4} = \frac{\log 5}{\log 2^2} = \frac{\log 5}{2\log 2}, (\because \log a^x = x \log a)$

$\therefore \log 2 = \frac{\log 5}{2a} \Rightarrow (1)$

$b = \frac{\log 6}{\log 5} = \frac{\log 2 + \log 3}{\log 5}, (\because \log(xy) = \log x + \log y)$

$\therefore \log 3 = b \log 5 - \log 2$

$\therefore \log 3 = b \log 5 - \frac{\log 5}{2a} = \frac{\log 5(2ab-1)}{2a} \Rightarrow (2)$

Since $\log 2 = \frac{\log 5}{2\text{a}}$

$\therefore \log_3 2 = \frac{\log 2}{\log 3} = \frac{\frac{\log 5}{2\text{a}}}{\frac{\log 5(2\text{ab}-1)}{2\text{a}}} = \frac{1}{2\text{ab}-1}$

Q.13 Answer:

0

Solution:

Given,

$3^{\log_4 5} - 5^{\log_4 3} \rightarrow \text{(i)}$

we know that

$a^{\log_c b} = b^{\log_c a}$

(1) becomes $3^{\log_4 5} - 3^{\log_4 5}$

=0

Hence Option A **is correct**

Q.14 Answer:

$(-4, -3) \cup (8, \infty)$

Solution:

Since $\frac{x^2+x}{x+4} > 0$ **and** $0 < 0.8 < 1$

$\therefore$ **the given inequality implies** $\log_6 \left(\frac{x^2+x}{x+4}\right) > 1$

$\Rightarrow \quad \frac{x^2+x}{x+4} > 6 \quad \Rightarrow \quad \frac{x^2-6x+x-24}{x+4} > 0 \quad \Rightarrow \quad x \in (-4, -3) \cup (8, \infty)$

Q.15 Answer:

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$2/3$

Solution:

Given, $y = \frac{3^x}{3} + \frac{1}{3^x.3}$

Now A.M $\geq$ **G.M** $\Rightarrow \frac{\frac{3^x}{3}+\frac{1}{3^x.3}}{2} \geq \sqrt{\frac{3^x}{3}\frac{1}{3^x.3}} \Rightarrow 3^{x-1} + 3^{-x-1} \geq \frac{2}{3}$

Q.16 Answer:

$(-\infty, -1]$

Solution:

Given that $x \in (-1, 3)\,(x - (a-1))\left(x - \left(a^2+2\right)\right) < 0 \quad \rightarrow \left(a^2+2\right) - (a-1) = a^2 - a + 3\ \Delta = (-1)^2 - 4 \times 3 = -11 < 0 \therefore \quad a^2 - a + 3 > 0\, a^2 + 2 > a - 1$ **(always)**

To make (1) negative, lower boundary of $x \geq \min\left\{a-1, a^2+2\right\}$ **and upper boundary of** $x \leq \max\left\{a-1, a^2+2\right\}$

$\therefore \quad a - 1 \leq -1$ **and** $a^2 + 2 \geq 3$

$a \leq 0$ **and** $\quad a^2 \geq 1$

$at(-\infty, 0]$ **and** $a\epsilon(-\infty, -1] \cup [1, \infty)$

$\therefore \quad$ **at** $(-\infty, -1]$

Q.17 Answer:

64

Solution:

$$\log_3(\log_2 x) + \log_{1/3}\left(\log_{1/2} y\right) = 1$$

$$\Rightarrow \log_3(\log_2 x) - \log_3\left(\log_{1/2} y\right) = 1$$

$$\Rightarrow \log_3\left(\log_2\left(4/y^2\right)\right) - \log_3\left(\log_{1/2} y\right) = 1$$

$$\Rightarrow \log_2\left(4/y^2\right) = 3\log_{1/2} y$$

$$\Rightarrow \log_2\left(4/y^2\right) = -3\log_2 y$$

$$\Rightarrow \log_2\left(4/y^2\right) + (\log_2 y)^3 = 0 \Rightarrow 4y = 1 \Rightarrow y = \frac{1}{4} \Rightarrow x = 64$$

Q.18 Answer:

2

Solution:

Given that

$(21.4)^a = (0.00214)^b = 100$

$(21.4)^a = 100, (0.00214)^b = 100$

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Applying log on both the sides, we get

$a = \log_{21.4} 100, b = \log_{0.00214} 100$

$\therefore \frac{1}{a} - \frac{1}{b} = \frac{1}{\log_{21.4} 100} - \frac{1}{\log_{0.00214} 100}, \because \left[\log_a b = \frac{1}{\log_b a}\right]$

$\Rightarrow \frac{1}{a} - \frac{1}{b} = \log_{100} 21.4 - \log_{100} 0.00214$

$\therefore \frac{1}{a} - \frac{1}{b} = \log_{100} \frac{21.4}{0.00214} = \log_{100} 100^2 = 2 \log_{100} 100 = 2,$

$\because \left[\log a - \log b = \log \frac{a}{b}\right]$

Q.19 Answer:

16

Solution:

$\log_2 \mathrm{x} + \log_2 \mathrm{y} \geq 6$

$\Rightarrow \mathrm{xy} \geq 64$

If $x > 0 \& y > 0$

Since A.M. $\geq$ **G.M**

$\Rightarrow \frac{x+y}{2} \geq \sqrt{xy}$

$\Rightarrow \frac{x+y}{2} \geq \sqrt{64}$ **. . . {From 1}**

$\Rightarrow x + y \geq 16$

Q.20 Answer:

$(11/3, \infty)$

Solution:

We have $||x-1|-5| < 2x - 5$

$\therefore 2x - 5 > 0$

$\Rightarrow x > \frac{5}{2}$

Now from (1)

$-2x + 5 < |x-1| - 5 < 2x - 5$

$\Rightarrow -2x + 10 < |x-1| < 2x$

Case I: $x \geq 1$

$\therefore -2x10 < |x-1| - 5 < 2x - 5$

$\Rightarrow x > 11/3$

Case II $x < 1$

$\therefore -2x + 10 < 1 - x < 2x$

$\Rightarrow x > 9$ and $x > 1/3$

$\Rightarrow x > 9$ **not possible as** $x < 1$

Therefore $x \in (11/3, \infty)$

Q.21 Answer:

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Solution:

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Let, $y = \frac{\log_2 24}{\log_{96} 2} - \frac{\log_2 192}{\log_{12} 2}$

$\Rightarrow y = (\log_2 24)(\log_2 96) - (\log_2 192)(\log_2 12), \left[\because \log_a b = \frac{1}{\log_b a}\right]$

$\Rightarrow y = (\log_2 2 + \log_2 12)(\log_2 8 + \log_2 12) -$

$(\log_2 12)(\log_2 16 + \log_2 12), [\because \log(ab) = \log a + \log b]$

$\Rightarrow y = (\log_2 2 + \log_2 12)\left(\log_2 2^3 + \log_2 12\right) - (\log_2 12)\left(\log_2 2^4 + \log_2 12\right)$

$\Rightarrow y = (1 + \log_2 12)(3 + \log_2 12) - (\log_2 12)(4 + \log_2 12), [\because \log_2 a = 1]$

$= 3 + \log_2 12 + 3\log_2 12 + \log_2 12 \cdot \log_2 12 - (4\log_2 12 + \log_2 12 \cdot \log_2 12) = 3$

$\Rightarrow y = 3$

Q.22 Answer:

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Solution:

$\log_3 \{5 + 4\log_3(x-1)\} = 2$

$\Rightarrow 5 + 4\log_3(x-1) = 3^2$

$\Rightarrow \log_3(x-1) = 1$

$\Rightarrow x - 1 = 3$

$\Rightarrow x = 4$

Q.23 Answer:

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Solution:

$\Rightarrow \sqrt{3}^{\left(-4+2\log_{\sqrt{5}} x\right)} = 9^{-1} = 3^{-2} = (\sqrt{3})^{-4}$

Equating the powers of $\sqrt{3}$

$-4 + 2\log_{\sqrt{5}} x = -4$

$\Rightarrow 2\log_{\sqrt{5}} x = 0$

$\Rightarrow \log_{\sqrt{5}} x = 0$

Taking exponential of the above equation, we get

$\therefore x = (\sqrt{5})^0 = 1$

Q.24 Answer:

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Solution:

$\log_2 x + \log_x 2 = \frac{10}{3} = \log_2 y + \log_y 2$ **...(1)**

$\text{and } x = y$

Equation (1) is valid when $x, y > 0$ **and** $x, y \neq 1$

$\log_2 x + \log_x 2 = \frac{10}{3}$

$\Rightarrow \log_2 x + \frac{1}{\log_2 x} = \frac{10}{3}, \left[\because \log_a b = \frac{1}{\log_b a}\right]$

On substituting $t = \log_2 x$ **in above equation, we get**

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$\Rightarrow 3t^2 - 10t + 3 = 0$
$\Rightarrow (3t - 1)(t - 3) = 0$
$\Rightarrow t = 3, \frac{1}{3}$
$\Rightarrow \log_2 x = 3, \frac{1}{3}$
$\Rightarrow x = 8, \sqrt[3]{2}$

Since, $x = y$

Therefore, $x + y = 16, 2\sqrt[3]{2}$

Q.25 Answer:

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Solution:

$N = 6\log_{10} 2 + \log_{10} 31$
$= \log_{10} 2^6 + \log_{10}(31)$
$= \log_{10}\left(2^6.31\right)$
$= \log_{10}(64 \times 31)$
$= \log_{10}(1984)$

Now $\log_{10}\left(10^3\right) = \log_{10}(1000) = 3$

Similarly $\log_{10}\left(10^4\right) = 4$

Since $10^3 < 1984 < 10^4$

$3 < \log_{10}(1984) < 4$

Hence it lies between 3 and 4. The sum of these two integers is 7.

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